Powering Tribal Resilience: Exploring Forest Biomass Energy Potential on Tribal Lands in the Pacific Northwest

April 2026

Research by: Jeremy Kwok Choon

EXECUTIVE SUMMARY

This report explores the intersection of forest stewardship and energy independence within the Pacific Northwest. Using GIS-based analysis, the study evaluates the potential for forest biomass (wood-based material removed during wildfire mitigation and forest restoration) to serve as a renewable energy resource for tribal nations in Oregon, Washington, and Idaho.

The analysis focuses on tribal land bases larger than 10,000 acres and incorporates a 5-mile buffer to account for nearby accessible resources. Key findings indicate a significant, though geographically concentrated, energy potential. The top 10 tribal land bases studied represent over 100 million Megagrams (Mg) of standing dry biomass, which is equivalent to approximately 549 million Megawatt-hours (MWh) of theoretical energy potential.

Recognizing that total extraction is neither ecologically nor logistically feasible, this study models annualized harvest cycles. Under a 40% thinning scenario, the top 10 tribes could collectively support between 5.5 million and 11 million MWh per year, depending on the rotation length. This resource scale illustrates how biomass can transition from a waste byproduct of forest management into a sustainable pillar of tribal energy sovereignty, providing local jobs and reducing wildfire risk through active stewardship.

While exploratory, these findings suggest that forest biomass is a viable regional energy resource. This report serves as a foundation for future, site-specific feasibility studies. These studies would include infrastructure assessments, road network analysis, and economic modeling to move from regional mapping toward community-led energy development.

1.0 INTRODUCTION

1.1 BACKGROUND

What if material removed to make forests safer could also mitigate wildfire risk, strengthen tribal energy sovereignty, and support rural jobs?

That question sits at the heart of this project.

Across the Pacific Northwest, many tribal communities are living with multiple, overlapping challenges. Forests in many places are denser than they once were, increasing wildfire risk. At the same time, many communities are seeking more local control over their energy future, especially as energy costs rise and climate-related disruptions become more common. In regions where traditional timber economies have declined and continue to decline, there is also growing interest in land-based economic opportunities that support both community well-being and forest stewardship.

This project explores one possible connection between these issues: **forest biomass.**

In simple terms, forest biomass is the wood-based material stored in trees and forest vegetation. When forests are thinned for timber production, forest restoration, or wildfire risk reduction, some of that material can potentially be used as an energy source. That means biomass may offer more than one benefit at the same time:

- It may help reduce hazardous fuel loads
- It may provide a renewable energy alternative
- It may support tribal energy sovereignty and resilience
- It may create work in forestry, hauling, processing, and energy systems

This study does **not** argue that every tribe should pursue biomass development, and it does **not** claim that every acre should be harvested. Instead, it asks a more basic and important first question:

1. Where does meaningful biomass energy potential exist near major tribal lands in Oregon, Washington, and Idaho?

That is the purpose of this analysis.

1.2 RESEARCH QUESTION

What is the forest bioenergy potential within 5 miles of major tribal reservations in the Pacific Northwest, and how does that potential change under different harvest rotation scenarios?

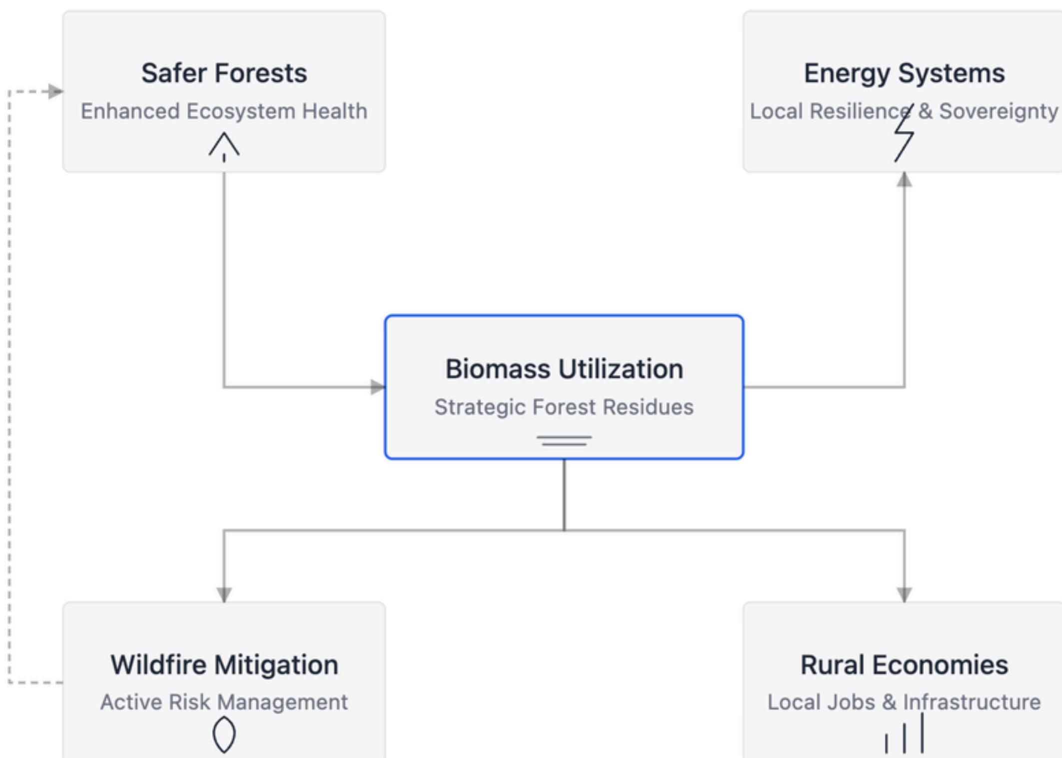
1.3 WHY THIS QUESTION MATTERS

This research is motivated by a practical and public-facing question, not just a technical one. For many people, “biomass” can sound abstract. But on the ground, this topic is really about community needs:

- safer forests
- more resilient energy systems
- more local control over energy resources
- more opportunities for rural and tribal economies

For tribal nations, this question can also connect to a larger issue of **energy sovereignty** — the ability to shape energy futures in ways that reflect local priorities, local resources, and local decision-making.

That is why this project matters. It is not only about how much biomass exists. It is about what kinds of possibilities that biomass might open up, such as energy resilience, wildfire mitigation, and promoting local economies.



Integrated Regional Resource Management

2.0 METHODOLOGY

2.1 STUDY AREA

To keep the analysis focused on larger land bases with greater potential for forest-based management, the study emphasizes tribal lands larger than 10,000 acres.

This project focuses on major tribal land bases in:

- Oregon
- Washington
- Idaho

A 5-mile buffer was added around tribal lands. This buffer is intended to serve as a reasonable starting point for considering nearby forest resources that may still be close enough to matter for hauling and local use. While tribes may not own all property in these reserves and surrounding reserves, they can harbour agreements with private or state entities through stewardship agreements or Good Neighbor Authority deals on Federal lands to steward these lands.

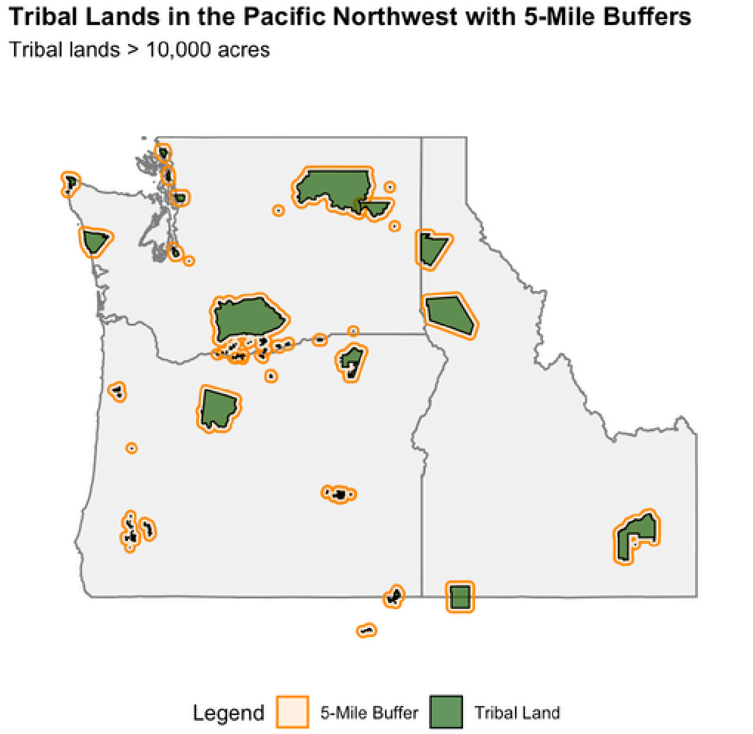
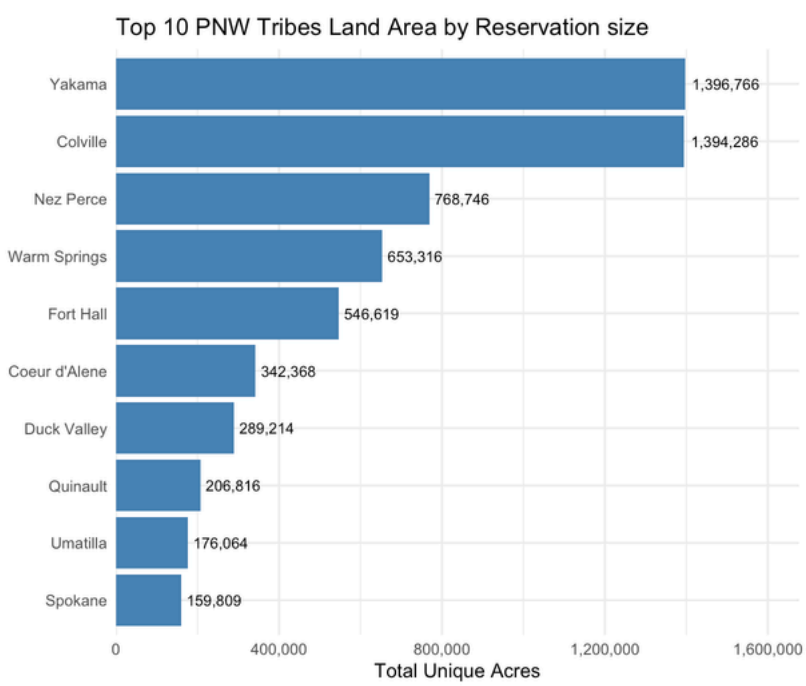


Figure 1: Tribal Lands with an area greater than 10,000 acres

2.2 STEP-BY-STEP: WHAT THIS PROJECT ACTUALLY DOES

This step-by-step breakdown of the methodology is for a general audience and should be able to follow the logic of the project in a few clear steps.

STEP 1: IDENTIFY TRIBAL LANDS



The first step was to map major tribal land areas in the Pacific Northwest. These lands are the center of the analysis. There are 43 recognized tribes in this region. The number of tribes was narrowed down to 10,000 acres or larger for the feasibility of biomass potential for the land area.

Figure 2: Top 10 PNW tribes by reservation size.

STEP 2: MEASURE FOREST STRUCTURE & CONDITIONS

To keep the analysis focused on larger land bases with greater potential for forest-based management, the study emphasizes tribal lands larger than 10,000 acres. Several forest datasets were used to understand what the surrounding landscape looks like. These include canopy height, canopy cover, canopy bulk density, forest type, and dry biomass.

These maps help answer basic questions such as:

- How much forest material is present?
- Where are the denser forested areas?
- Which places appear to have larger wood-based resource potential?

Data was collected from LANDFIRE, whose principal partners are the Department of the Interior (DOI) & USDA Forest Service (USFS)

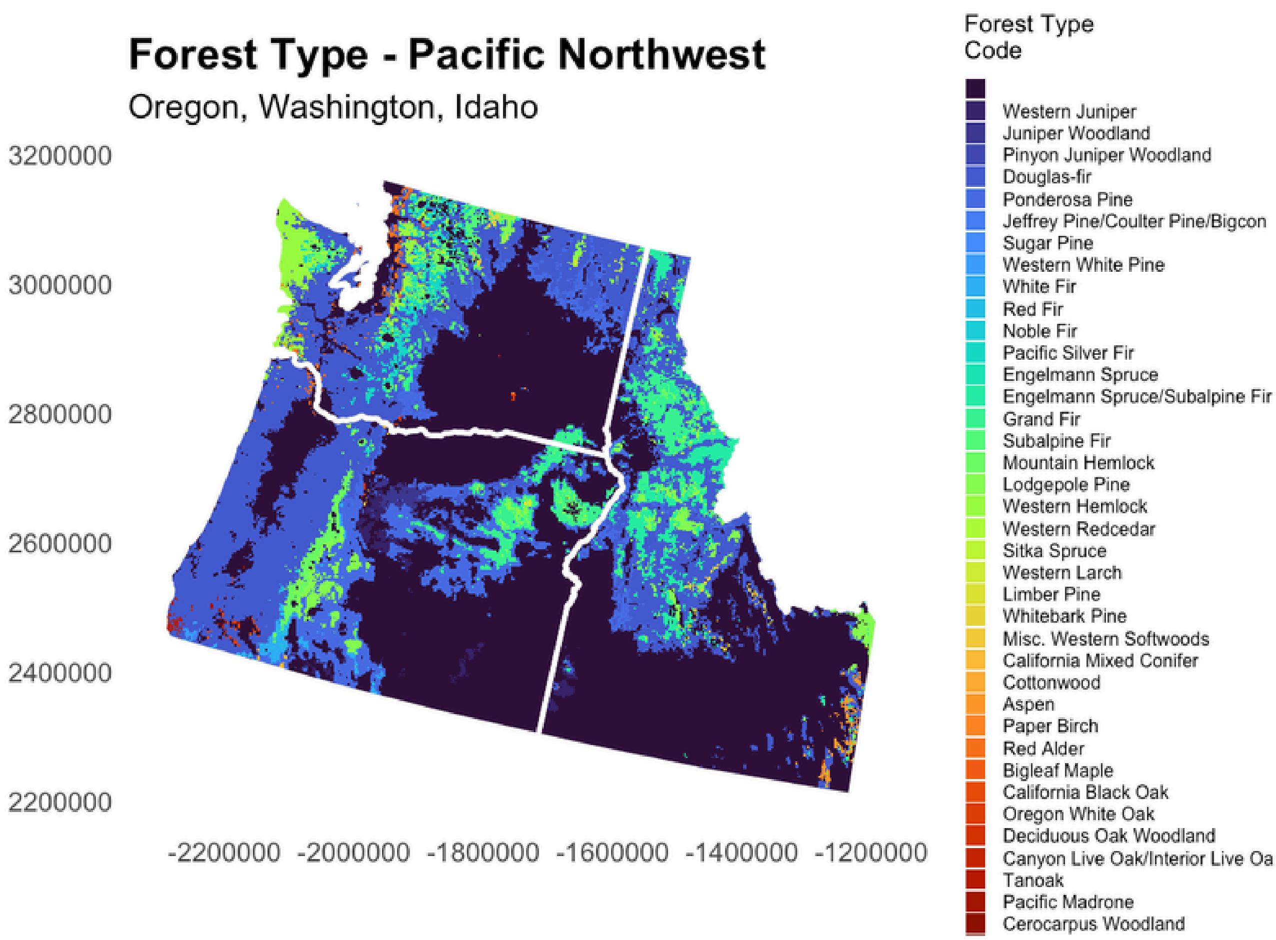


Figure 3: Forest type distribution across the Pacific Northwest.

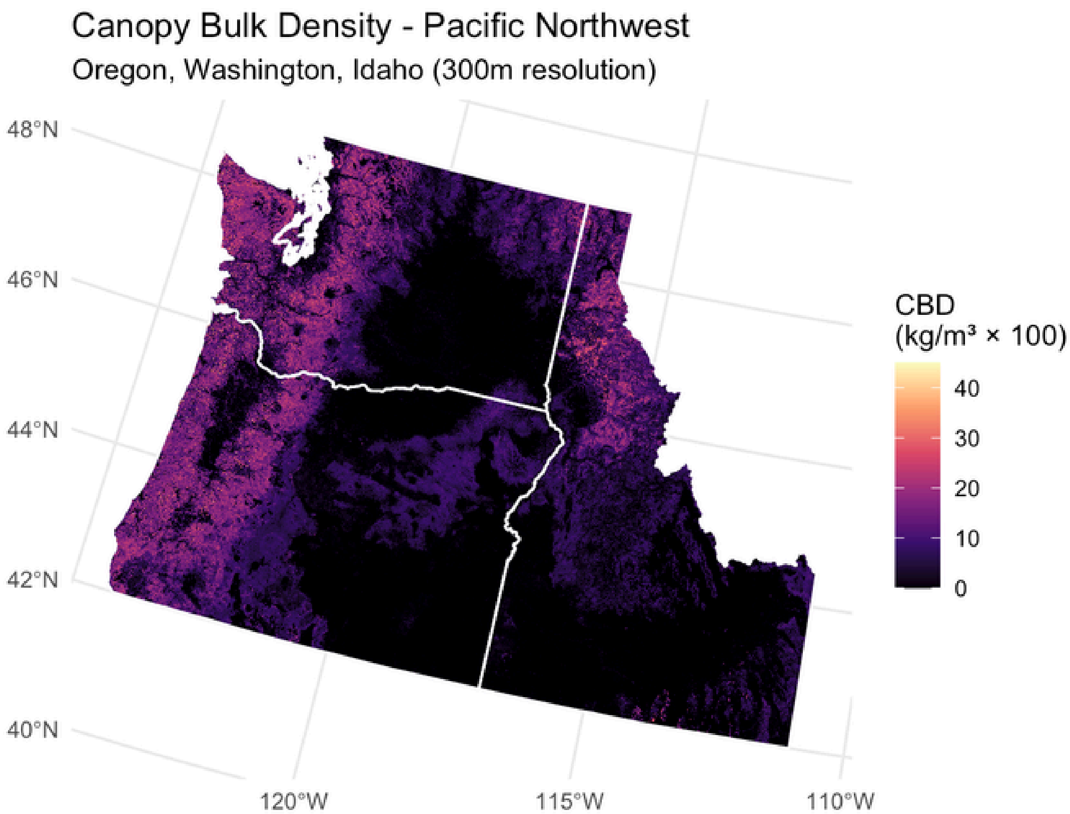


Figure 4: Canopy bulk density across the Pacific Northwest.(Canopy Bulk Density (CBD) is the mass of available canopy fuel (foliage and small branches) per unit of canopy volume, typically measured in. It measures how closely fuels are packed in the forest canopy, serving as a critical indicator for modeling the likelihood and intensity of active crown fires.)

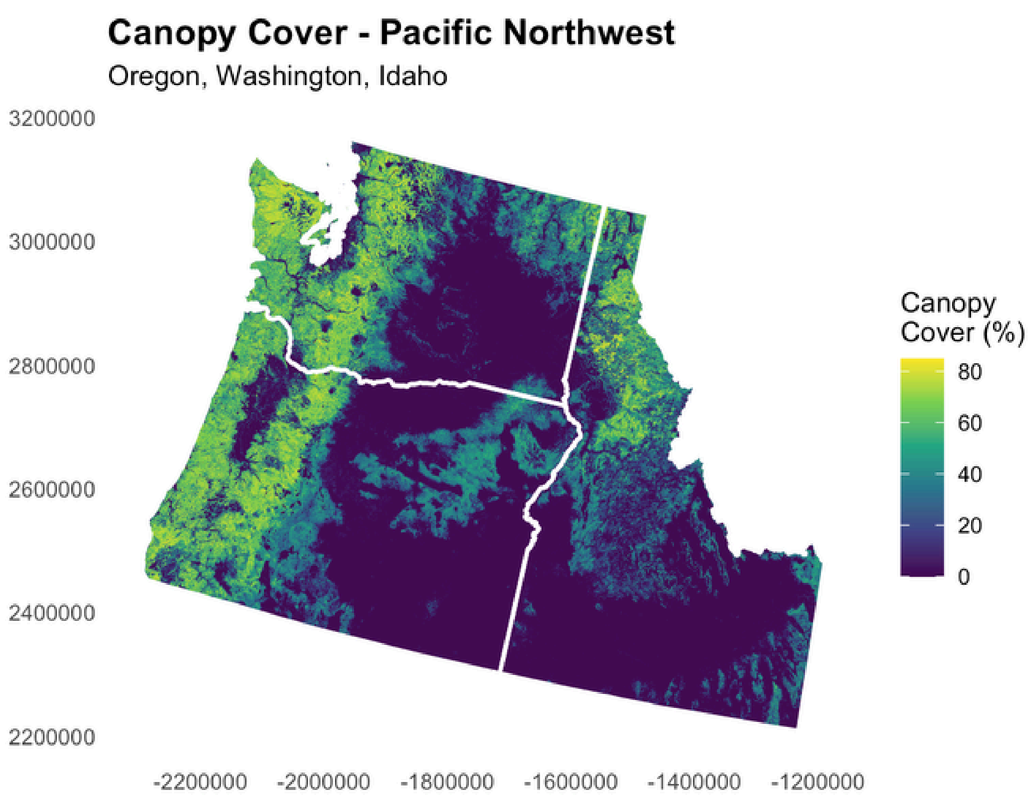


Figure 5: Canopy cover across the Pacific Northwest.

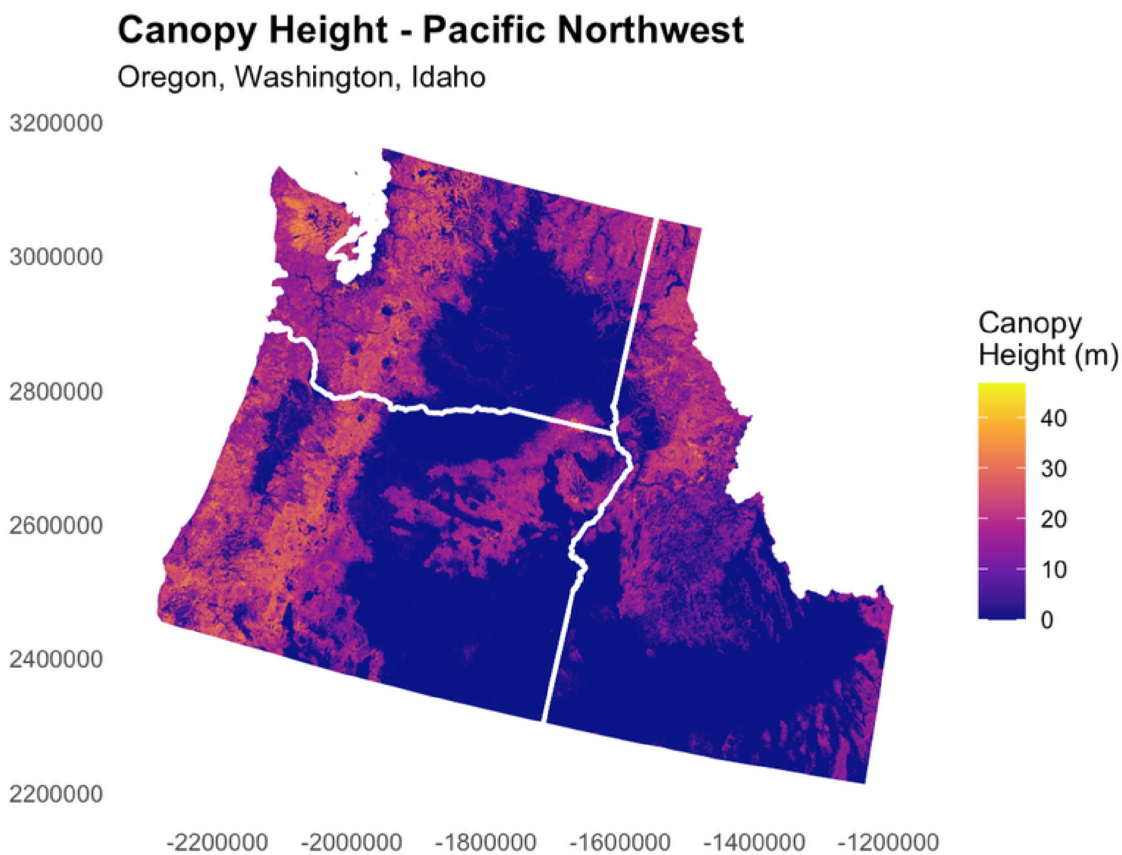


Figure 6: Canopy height across the Pacific Northwest.

These maps themselves do not calculate biomass, but they were used to create a biomass layer used in the next step. The Biomass layer in the following Step was created by Landfire.

STEP 3: MEASURE DRY BIOMASS

The key resource variable in this study is dry biomass.

Dry biomass is the estimated amount of dry wood material above ground. In this project, it is measured in Mg, or megagrams.

I used the USDA Forest Service FIA/MODIS Biomass dataset. This is a 250-meter resolution model that correlates on-the-ground measurements from ~500,000 FIA sample plots with satellite imagery (MODIS), climate data, and topographic layers. It provides a spatially continuous estimate of forest biomass that is statistically anchored to real-world plot measurements.

A simple way to think about it is:

1 Mg = 1 metric ton

- So when this report talks about millions of Mg, it is talking about millions of metric tons of dry wood material.

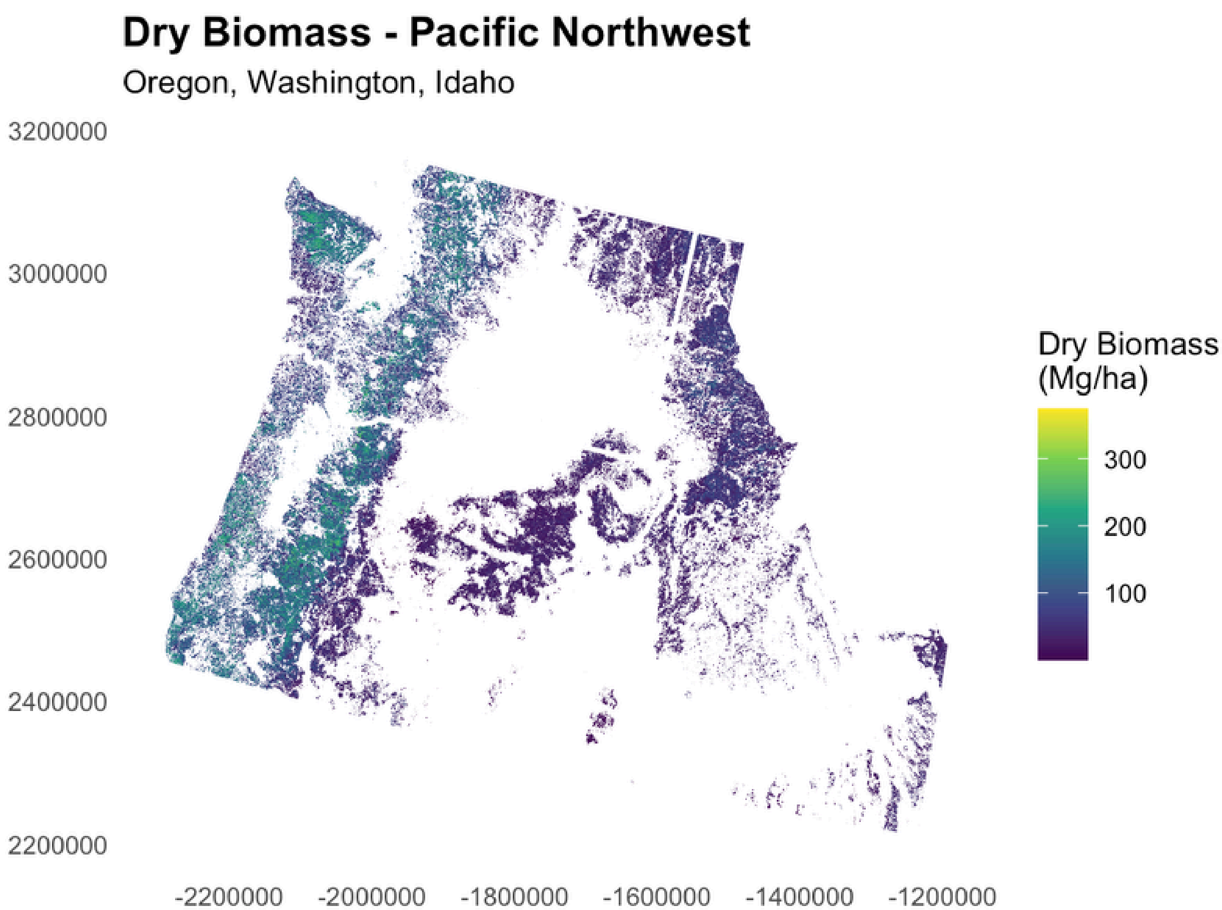


Figure 7. Dry biomass distribution across the Pacific Northwest.

STEP 4: OVERLAY BIOMASS WITH TRIBAL LANDS AND BUFFERS

After mapping biomass, the study overlays that information with tribal lands and their 5-mile buffers.

This helps identify where tribal geographies and forest biomass resources overlap.

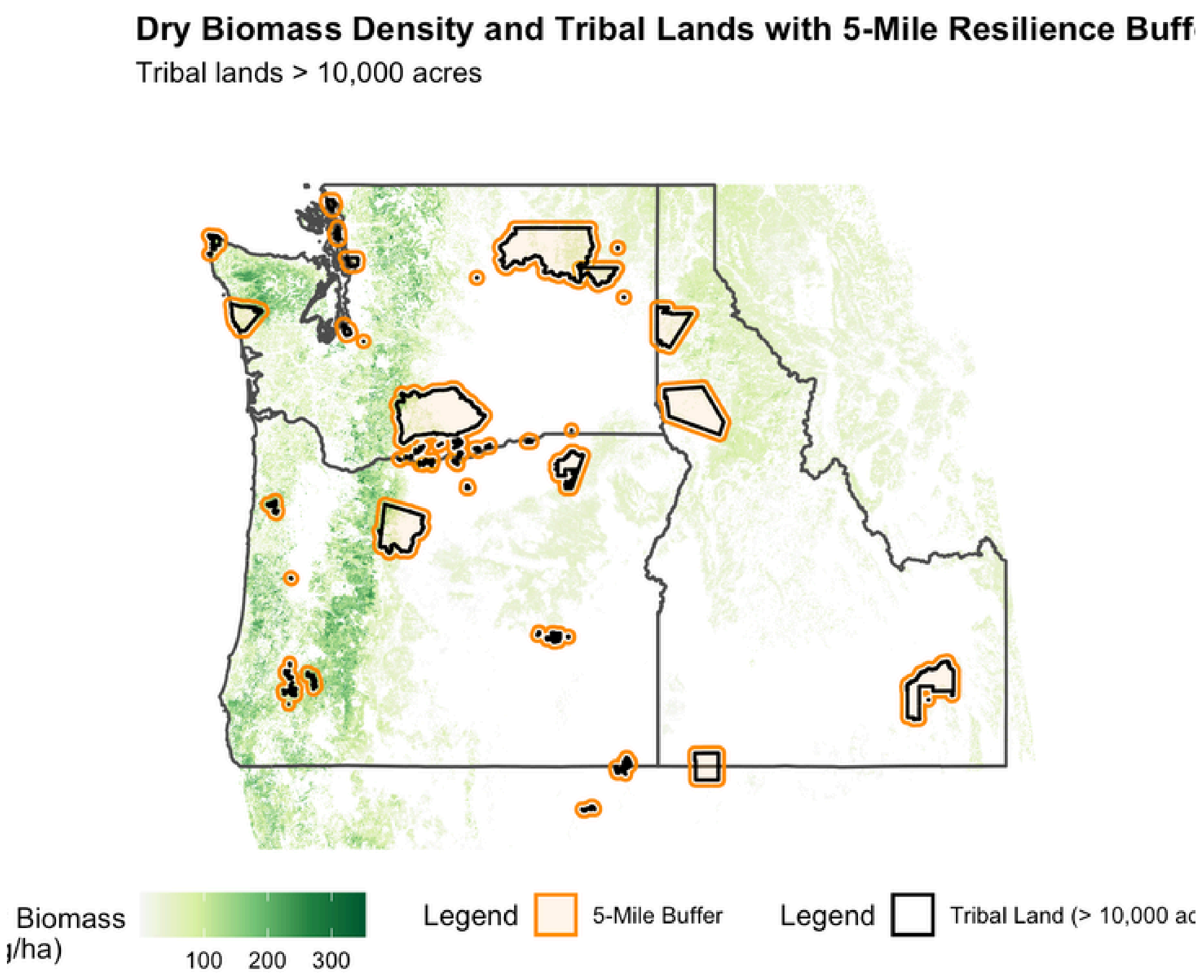


Figure 8. Biomass density in relation to tribal lands and the 5-mile buffer.

STEP 5: CONVERT BIOMASS INTO ENERGY POTENTIAL

Once the biomass amount is estimated, it can be translated into a potential energy value. This does not mean all of the biomass should or would be used. It simply gives a way to understand the scale of the resource.

STEP 6: TEST HARVEST SCENARIOS

Finally, the study tests different scenarios for how biomass might be spread over time. Instead of assuming all material is removed at once, the project looks at annualized harvest cycles. This makes the results easier to interpret as a long-term resource rather than a one-time extraction.

3.0 THE MATH BEHIND THE CALCULATIONS

One goal of this project is to make the math understandable, even for readers who do not work with GIS, forestry or energy data.

3.1 FROM BIOMASS TO ENERGY

The study uses a Higher Heating Value (HHV) assumption of:

$$19.0\ \mathrm{GJ/Mg}$$

This means each megagram of dry biomass is assumed to contain about 19 gigajoules of energy.

To express that in megawatt-hours, the study uses the conversion:

$$1\ \mathrm{GJ} = 0.277778\ \mathrm{MWh}$$

So the energy per megagram becomes:

$$19.0 \times 0.277778 \approx 5.28\ \text{MWh per Mg}$$

That leads to the basic formula:

$$\text{Total Energy (MWh)} = \text{Dry Biomass (Mg)} \times 5.28$$

3.2 FROM ENERGY TO HOUSEHOLD EQUIVALENTS

To make the results easier for the public to understand, the study also converts energy into an estimated number of homes.

The project uses an average annual household electricity use of about:

$$10.8\ \text{MWh per home per year}$$

So:

$$\text{Homes Powered} = \text{Energy (MWh)}/10.8$$

This gives a rough “household equivalent” that helps communicate scale.

3.3 THINNING SCENARIO MATH

Thinning scenarios are critical because they translate a static snapshot of forest biomass into a realistic, annualized energy supply. In energy planning, knowing the total volume of wood on the landscape is less useful than knowing how much can be sustainably harvested and utilized each year. The study also models a 40% thinning scenario. This means the analysis assumes that only 40% of the total biomass resource is removed within a given management cycle.

That amount is then spread across different rotation lengths:

- 20 years
- 30 years
- 40 years

The annual energy formula is:

$$\text{Annual Energy (MWh/year)} = \frac{\text{Total Energy} \times 0.40}{\text{Rotation Length}}$$

And annual household equivalents are calculated as:

$$\text{Annual Homes Powered} = \frac{\text{Annual Energy}}{10.8}$$

EXAMPLE: YAKAMA

For Yakama, the total theoretical energy potential shown in the results table is:

$$111{,}113{,}567 \text{ MWh}$$

Under the 40% thinning scenario and a 20-year cycle:

$$\frac{111{,}113{,}567 \times 0.40}{20} = 2{,}222{,}271 \text{ MWh/year}$$

Then the household equivalent becomes:

$$\frac{2{,}222{,}271}{10.8} \approx 205{,}766 \text{ homes}$$

This is the logic behind the scenario tables in the results.

4.0 FINDINGS

4.1 TOTAL THEORETICAL RESOURCE FOR THE TOP 10 TRIBES

The top 10 tribal land bases in the study account for:

- **104,181,929 Mg** of standing dry biomass
- **549,849,508 MWh** of total theoretical energy potential
- **50,911,993 household** equivalents as a scale comparison

These numbers are large, but they should be understood carefully. They describe the overall scale of the resource, not what should be harvested all at once.

Top 10 PNW Tribal Biomass Energy Potential

Tribe	Area (Acres)	Dry Biomass (Mg)	Energy (MWh)	Homes Powered
Yakama	2,840,441	21,053,080	111,113,567	10,288,293
Warm Springs	1,503,611	16,861,236	88,989,928	8,239,808
Cow Creek	575,262	16,619,214	87,712,589	8,121,536
Colville	2,277,333	13,682,161	72,211,463	6,686,247
Quinault	508,202	12,417,718	65,538,009	6,068,334
Grand Ronde	231,222	6,518,314	34,402,240	3,185,393
Coeur d'Alene	734,859	5,654,514	29,843,290	2,763,268
Nez Perce	1,283,224	4,412,518	23,288,305	2,156,325
Spokane	587,749	4,004,898	21,136,980	1,957,128
Makah	237,298	2,958,276	15,613,137	1,445,661

Figure 9. Top 10 tribes by total biomass energy potential.

4.2 ENERGY POTENTIAL IS NOT EVENLY DISTRIBUTED

The results show that biomass energy potential is concentrated in a relatively small number of places.

The leading tribes by total potential are:

- Yakama
- Warm Springs
- Cow Creek
- Colville
- Quinault

Together, these five account for the large majority of the top-10 total. This matters because it suggests that future biomass conversations may be most relevant in specific geographic hotspots, rather than evenly across the entire region.

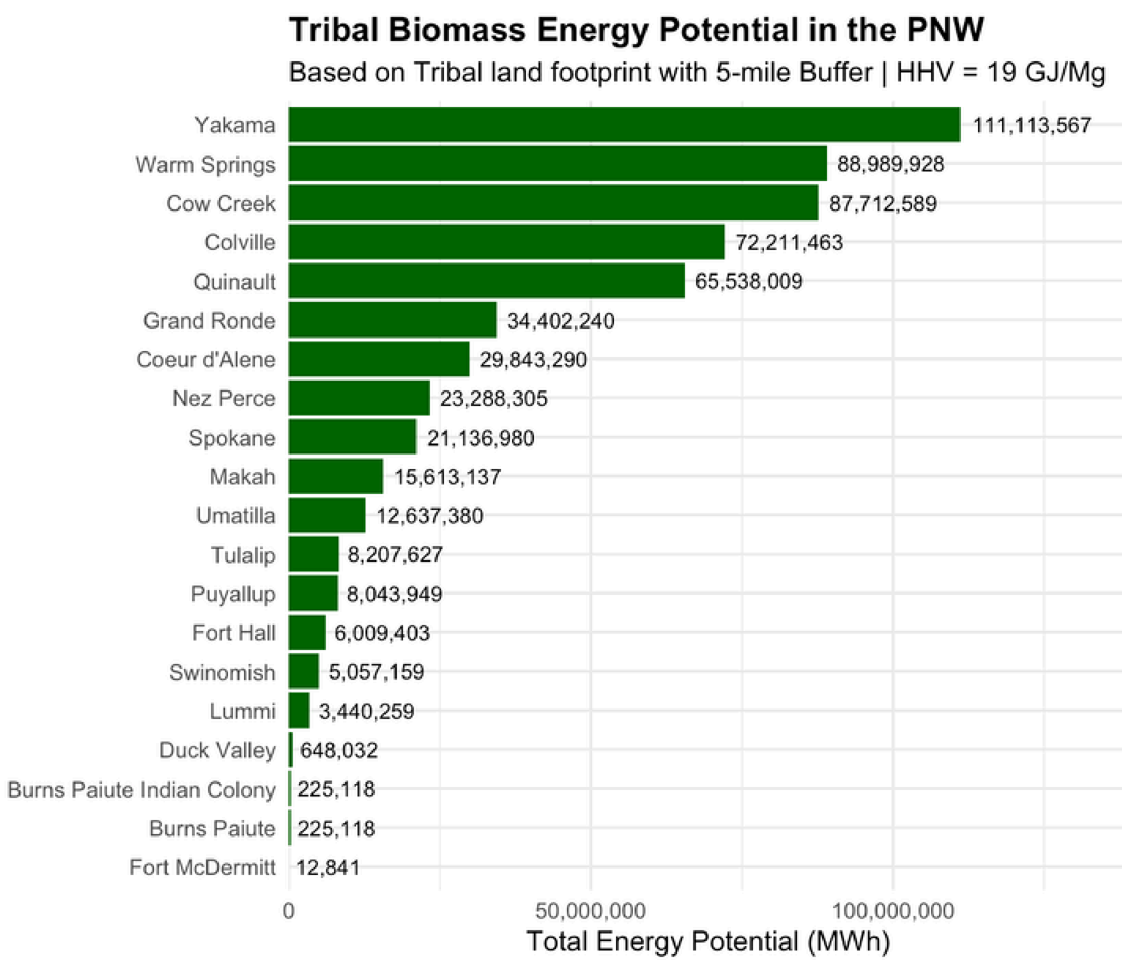


Figure 10. Total theoretical biomass energy potential by tribe.

4.3 ANNUAL SCENARIOS TELL A MORE REALISTIC STORY

A more useful public question is not “How much energy exists in total?” but:

How much energy could be available each year under a more realistic management approach?

That is where the thinning scenarios matter. Under a 40% thinning scenario, the top 10 tribes together could support approximately:

- 10,996,991 MWh/year on a 20-year cycle
- 7,331,327 MWh/year on a 30-year cycle
- 5,498,495 MWh/year on a 40-year cycle

In household-equivalent terms, that is about:

- 1,018,240 homes/year on a 20-year cycle
- 678,826 homes/year on a 30-year cycle
- 509,119 homes/year on a 40-year cycle

These are still large numbers, but they tell a more grounded story than the total resource estimate. They show how biomass might function as a long-term regional energy resource under a management scenario, rather than as a one-time stockpile.

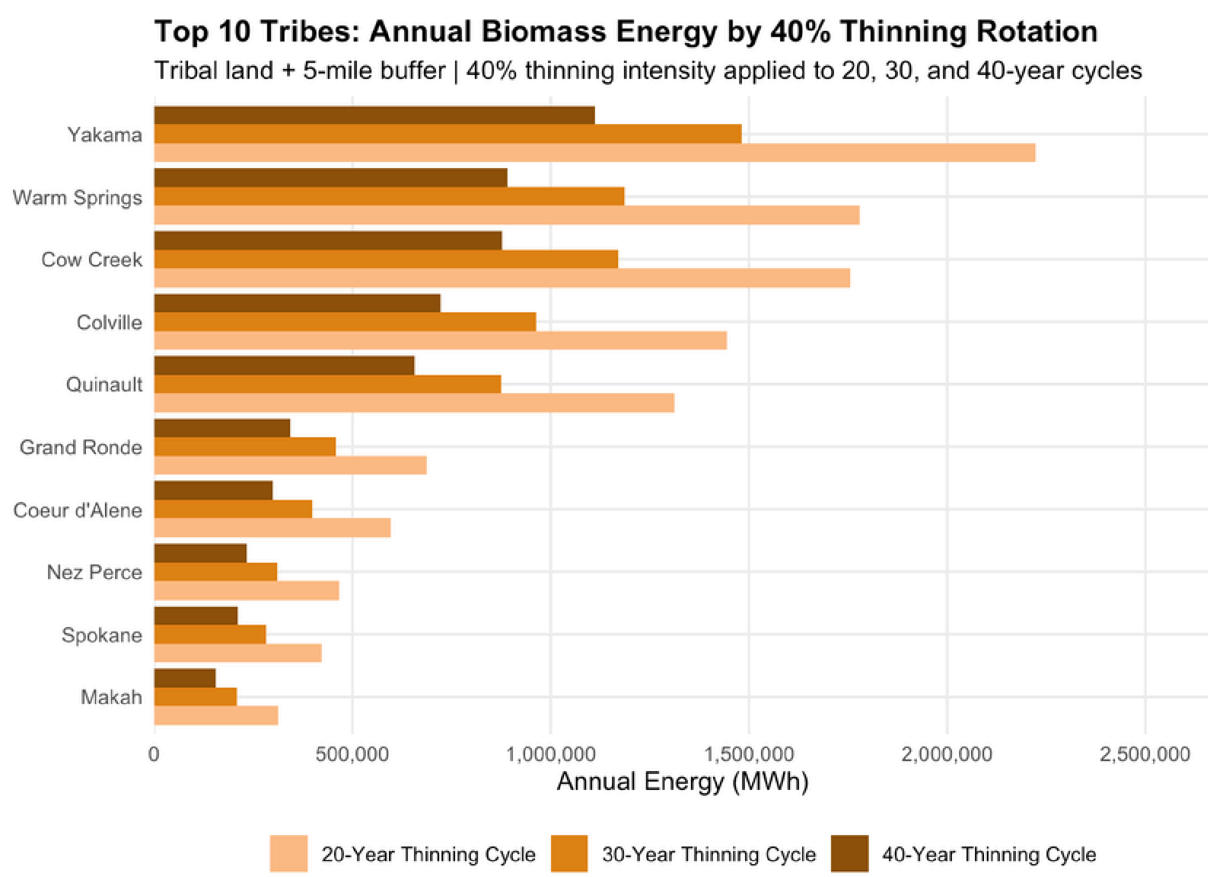


Figure 11. Annual biomass energy by tribe under 40% thinning scenarios.

Thinning Harvest Scenarios (40% Intensity) by Rotation Cycle — Tribal Land + 5-Mile Buffer

	20-Year Thinning Cycle		30-Year Thinning Cycle		40-Year Thinning Cycle	
Tribe	**Annual MWh**	**Homes**	**Annual MWh**	**Homes**	**Annual MWh**	**Homes**
Yakama	2,222,271	205,766	1,481,514	137,177	1,111,136	102,883
Warm Springs	1,779,799	164,796	1,186,532	109,864	889,899	82,398
Cow Creek	1,754,252	162,431	1,169,501	108,287	877,126	81,215
Colville	1,444,229	133,725	962,820	89,150	722,115	66,862
Quinault	1,310,760	121,367	873,840	80,911	655,380	60,683
Grand Ronde	688,045	63,708	458,697	42,472	344,022	31,854
Coeur d'Alene	596,866	55,265	397,911	36,844	298,433	27,633
Nez Perce	465,766	43,126	310,511	28,751	232,883	21,563
Spokane	422,740	39,143	281,826	26,095	211,370	19,571
Makah	312,263	28,913	208,175	19,275	156,131	14,457

Figure 12. Annual thinning scenario results for the top 10 tribes with a 5 mile buffer with 40% intensity

5.0 WHY THESE FINDINGS MATTER

5.1 BIOMASS COULD SUPPORT ENERGY SOVEREIGNTY

For tribal communities, local energy systems can matter for more than economics. They can also matter for self-determination, resilience, and long-term planning.

Biomass has the potential to become part of a broader energy sovereignty strategy when it is community-led, locally appropriate, and aligned with tribal priorities.

In plain language, this means energy produced closer to home, using local resources, with more local control over the benefits.

5.2 BIOMASS COULD SUPPORT WILDFIRE MITIGATION

In many places, hazardous fuels and dense forest conditions increase wildfire risk.

That means some of the same material that poses a fire hazard may also represent an energy opportunity. If done carefully, thinning for forest health and thinning for biomass use do not have to be competing ideas. They can be part of the same conversation.

This does not mean biomass is a cure-all for wildfire. But it does suggest that restoration-oriented thinning could provide multiple benefits at once.

5.3 BIOMASS COULD SUPPORT RURAL AND TRIBAL JOBS

Many forest-based regions have experienced long-term changes in timber markets and wood-processing infrastructure.

Biomass will not replace the historical wood products economy, but it may offer a partial and place-based opportunity in some communities. That could include jobs and contracting opportunities in areas such as:

- forest restoration
- fuels reduction
- hauling and logistics
- biomass processing
- facility operations
- local energy planning

For that reason, biomass should be understood not only as an energy topic, but also as part of a wider discussion about rural development and land-based economic opportunity.

6.0 WHAT THIS STUDY DOES NOT YET ANSWER

6.1 LIMITATIONS

This project is **exploratory.**

That is one of its strengths, because it opens the conversation. But it also means the results should be interpreted with care.

Several important real-world factors are not yet built into this analysis:

- **Wood type and species:** not all biomass has the same value or usability
- **Slope and terrain:** Some wood may exist on land that is difficult or expensive to access
- **Road infrastructure:** transport depends on road quality, location, and distance
- **Extraction costs: removing and moving material costs money**
- **Conversion technology:** actual delivered electricity would depend on plant efficiency and system design
- **Policy and permitting:** energy projects depend on regulation, governance, and local priorities
- **Cultural and ecological considerations:** tribal decisions about land and energy must reflect more than resource quantity alone

This is why the project should be seen as a conversation starter, not a final development blueprint.

6.2 IMPORTANT INTERPRETATION NOTE

The energy estimates in this project are best understood as resource potential or energy-equivalent estimates based on biomass fuel value.

In other words:

- They show the scale of energy contained in the biomass resource
- They do not yet model full real-world conversion losses, facility design, or delivered electric output

That means actual electricity production from a biomass facility would likely be lower than the raw energy-equivalent estimates shown here.
This is an important next step for future research.

6.3 WHERE THIS RESEARCH CAN GO NEXT

This analysis helps answer the first question: Where is the resource, and how large is it?

The next phase would ask more practical questions:

- Which sites are physically feasible?
- Which locations have enough nearby biomass to support a facility?
- Where are roads, mills, or transmission lines already present?
- What are the likely extraction and hauling costs?
- What technologies are most appropriate for different tribal contexts?
- How do energy opportunities align with tribal priorities for stewardship, sovereignty, and development?

Future work could build on this study by adding:

- road network analysis
- slope and accessibility analysis
- facility siting analysis
- biomass transport cost modeling
- conversion efficiency estimates
- tribal energy demand analysis
- community-specific case studies

That is where this research could become even more useful: not simply as a regional map of possibility, but as a tool for place-based planning.

6.4 FINAL REFLECTION

At its core, this project is about possibility.

It asks whether material often treated as waste, hazard, or byproduct might instead be part of a more resilient future.

It suggests that in some places, forest biomass may offer a bridge between:

- wildfire mitigation
- renewable energy
- tribal energy sovereignty
- rural economic opportunity

Not every place will be suitable. Not every project will make sense. And not every biomass pathway will be worth pursuing.

But this analysis shows that the conversation is worth having.

The resource is there. The geography matters. The opportunities are uneven but real. And with deeper analysis, stronger partnerships, and community-led decision-making, this work can move from exploratory mapping toward practical planning.

Contact Information

Jeremy Kwok Choon
Github: https://github.com/JerbearKCodes
LinkedIn: www.linkedin.com/in/jeremy-kwok-choon